INCH-POUND

MIL-C-50652/5A AR)
12 December 1990
SUPERSEDING
MIL-C-50652/5(MU)
15 April 1971

# MILITARY SPECIFICATION

## CIRCUIT CARD ASSEMBLY: 10548016

## (GENERATOR, RCFD)

**This specification Is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.**

## 1. SCOPE

**1.1 Scope.** The detail requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the Range Control Fine Delay Generator (RCFD Generator) used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. This assembly provides vernier delays of ± 1.25 meters of the E/L coarse trigger produced in the range counter assembly.

## 2. APPLICABLE DOCUMENTS

**2.1 Government documents.**

**2.1.1 Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

**SPECIFICATIONS**

**MILITARY**

MIL-C-50652 - Circuit Card Assemblies, General Specification for

Beneficial comments, recommendations, additiions, deletions) and any pcrtinent data which may bc of use in improving this documment should be adressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BACS, Picaninny Arsenal, New Jersey 07806-5000 by using the self—addressed Standardization Document Imporvement Proposal DD Form 1426) appearing at the end of this document or by letter.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other **Government documents, drawings and publications.** The following other Government documents, drawings **and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on** the **date of the solicitation.**

**DRAWINGS**

**U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)**

**F10548O16 - Assembly, RCFD** Generator

**SPECIAL PACKAGING INSTRUCTION**

**PS10548O16 - packaging of RCFD Generator**

**(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)**

**2.3** Order of precedence. **In** the **event** of conflict **between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.**

*3.* **REQUIREMENTS**

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

**3.2 General specifications. The assembly shall meet the requirements of MIL-C-50652.**

**3.2.1 Components and assemblies. The components and assemblies shall** comply with all requirements specified on Drawing F10548O16, all associated drawings, and with all requirements specified in applicable specifications and standards.

**3.3 Environmental service conditions. As specified in** MIL-C-**50652.**

3.4 **Electrical. Each conductor shall be electrically continuous as specified on** the **schematic diagram** of the circuit card **assembly.**

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3*5 Performance. When supplied with the input signals and logic levels specified in Table I and Table 11 respectively the RCFD generator shall yield the output signal with delays specified in Table I and Table 11 respectively. All delays shall be measured at the 50 *percent* points.

TABLE I. Input and output characteristics

| REQUIREMENT | CHARACTERISTICS | TERMINALS |
|---|---|---|
| Input power | 5.0 ± 0.3 Vdc regulated; ac ripple less than 50 mV peak-to-peak, capacity 300 mA | P1-30, P1-31, & P1-32 (+5 V) P1-24, P1-25, & P1-26 (5 V RET) |
| | 25 ± 0.5 Vdc regulated; ac ripple less than 50 mV peak-to-peak; capacity 100 mA | P-55 & P1-56 (+25 V) P1-53 & P1-54 (25 V RET) |
| Input signal | 20 kHz positive pulse of not less than 3 Vdc amplitude, 0.4 us wide. | P1-51 |
| Input logic levels | See Table II. | |
| Output signal | 20 kHz positive pulse width; "O" state: 0.0 ± 0.5V; "l" state: 3.5 ± l.OV; 0.4 to 1*5 us wide at 50 percent points; rise time between 10 percent and 90 percent points is 20 ns max. See Table II for delay measurements. | P1-11 & P1-41 |

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TABLE II. SIGNAL DELAY LIMITS

| STEP | LOGIC 1 TERMINALS | INPUT SWITCH | OUTPUT TERMINALS | TEST POINT | DELAY LIMITS (ns) NLT | NMT | NOTES |
|---|---|---|---|---|---|---|---|
| 1 | P1-48 | 1 | P1-41 | 1 | 30.0 | 75.0 | 1, 3 |
| 2 | P1-45 | 2 | P1-41 | 1 | 6.8 | 9.6 | 2, 3 |
| 3 | P1-39 | 3 | P1-41 | 1 | 6.8 | 9.6 | 2, 3 |
| 4 | P1-35 | 4 | P1-41 | 1 | 6.8 | 9.6 | 2, 3 |
| 5 | P1-47 | 5 | P1-41 | 1 | 6.8 | 9.6 | 2, 3 |
| 6 | P1-43 | 6 | P1-41 | 1 | 6.8 | 9.6 | 2, 3 |
| 7 | P1-37 | 7 | P1-41 | 1 | 6.8 | 9.6 | 2, 3 |
| 8 | P1-23 | 8 | P1-41 | 1 | 608 | 9.6 | 2, 3 |
| 9 | P1-21 | 9 | P1-11 | 2 | 6.8 | 9.6 | 2, 3 |
| 10 | P1-15 | 10 | P1-11 | 2 | 6.8 | 9.6 | 2, 3 |
| 11 | P1-9 | 11 | P1-11 | 2 | 6.8 | 9.6 | 2, 3 |
| 12 | P1-5 | 12 | P1-11 | 2 | 608 | 9.6 | 2, 3 |
| 13 | P1-17 | 13 | P1-11 | 2 | 6.8 | 9.6 | 2, 3 |
| 14 | P1-13 | 14 | P1-11 | 2 | 6.8 | 9.6 | 2, 3 |
| 15 | P1-7 | 15 | P1-11 | 2 | 6.8 | 9.6 | 2, 3 |
| 16 | P1-3 | 16 | P1-11 | 2 | 6.8 | 9.6 | 2, 3 |

NOTES: 1. Delay is from leading edge of input signal T1 specified in Table I to leading edge of output signal specified in Table I.

2. Delay is from leading edge of output signal of previous step and step being presently observed. The cumulative delay from first to last leading edge shall be 123 ± 9ns.

3. Switch as specified for application of logic 1 shall be in position 1 and all others in position 2 for logic 0.

3.6 First article. As specified in MIL-C-50652.

3.7 Soldering. As specified in MIL-C-50652.

4* QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification of the circuit card assembly.

4.2 First article inspection. As specified in MIL-C-50652.

4.3 Inspection provisions. As specified in MIL-C-50652.

4*4 Special sampling. As specified in MIL-C-50652.

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4.5 Inspection equipment. As specified in MIL-C-50652 and 4.5.3.

4.5.1 Government-furnished Inspection equipment. As specified in MIL-C-50652.

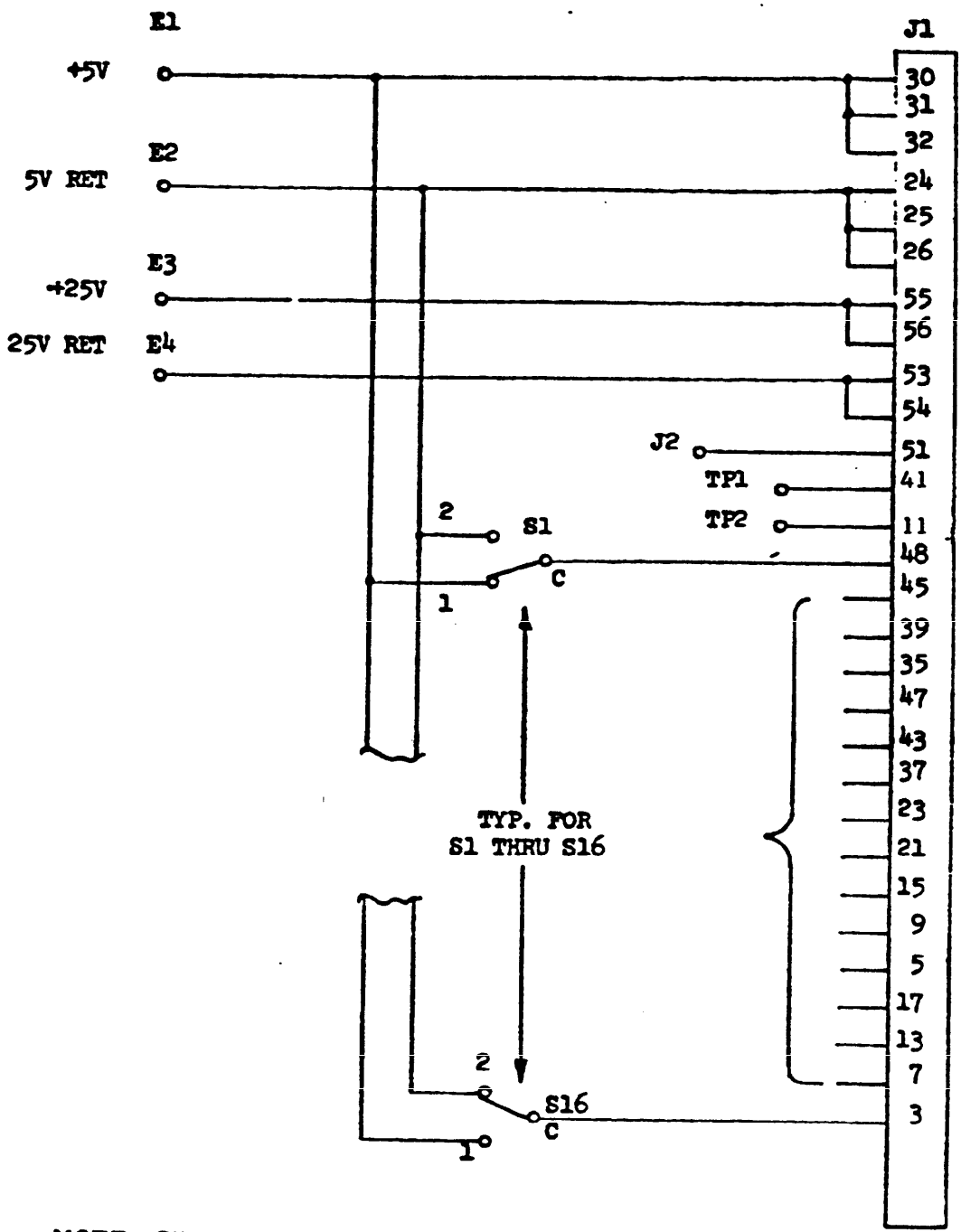
4.5.2 Contractor-furnished inspection equipment. As specified in MIL-C-50652.

4.5.3 Test equipment. In conjunction with 4.5.1 and 4.5.2 the test equipment listed in Table III shall be utilized in the performance **of the applicable tests as specified in 4.6.3.**

TABLE III. TEST EQUIPMENT LIST

| ITEM NUMBER | DESCRIPTION |
|---|---|
| 1. Power Supply | **5.0 ± 0.3 Vdc regulated; ac ripples less than 50 mv peak-to-peak; capacity 300 mA.** |
| 2. Power Supply | **25 ± 0.5** Vdc regulated; ac ripple less than 50 mv peak-to-peak; capacity 100 mA. |
| **3. Pulse generator** | **20 kHz positive pulse, amplitude 3 Vdc peak-to-peak, into 50-ohm load.** |
| 4. ***Test* fixture** | See Figure 1. See Table IV for test fixture parts list. |
| 5* Oscilloscope | Dual charnel; rise time of scope plus measuring probe 2 ns or less, sweep period 1 ns per division or less, input resistance greater than 8 megohms, input capacitance less than 25 pF. |

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NOTE: SEE TABLE IV FOR TEST FIXTURE PARTS LIST

FIGURE 1. Test fixture setup

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TABLE IV. TEST FIXTURE PARTS LIST

| DESIGNATION | NOMENCLATURE AND DESCRIPTION |
|---|---|
| El | Binding post, H.H. Smith 257-R or equivalent. |
| E2 | Same as El. |
| E3 | Binding post, H.H. Smith 257-B or equivalent. |
| E4 | Same as E3. |
| J1 | Connector; any of the following or equivalent:<br>(1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois.<br>(2) 500946-01, National Connector Corp. Minneapolis, Mire.<br>(3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa. |
| J2 | Connector, receptacle, UG-25B/U or equivalent. |
| S1 thru S16 | Switch SPDT. |
| TPl | Tip jack, H.H. Smith 1501 or equivalent. |
| TP2 | Same as TP1. |

4.6 Methods of Inspection.

4.6.1 Environmental. As specified in MIL-C-50652.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.6.3 Performance. The selected RCFD generator shall be tested as follows:

4.6.3.1 Test setup. Adjust power supplies and pulse generator, items 1 through 3, to deliver outputs described as input power and input signals in Table I. Connect test equipment, items 1 through 5, as shown in Figure 2. Insert RCFD generator into J1 in test fixture item 4.

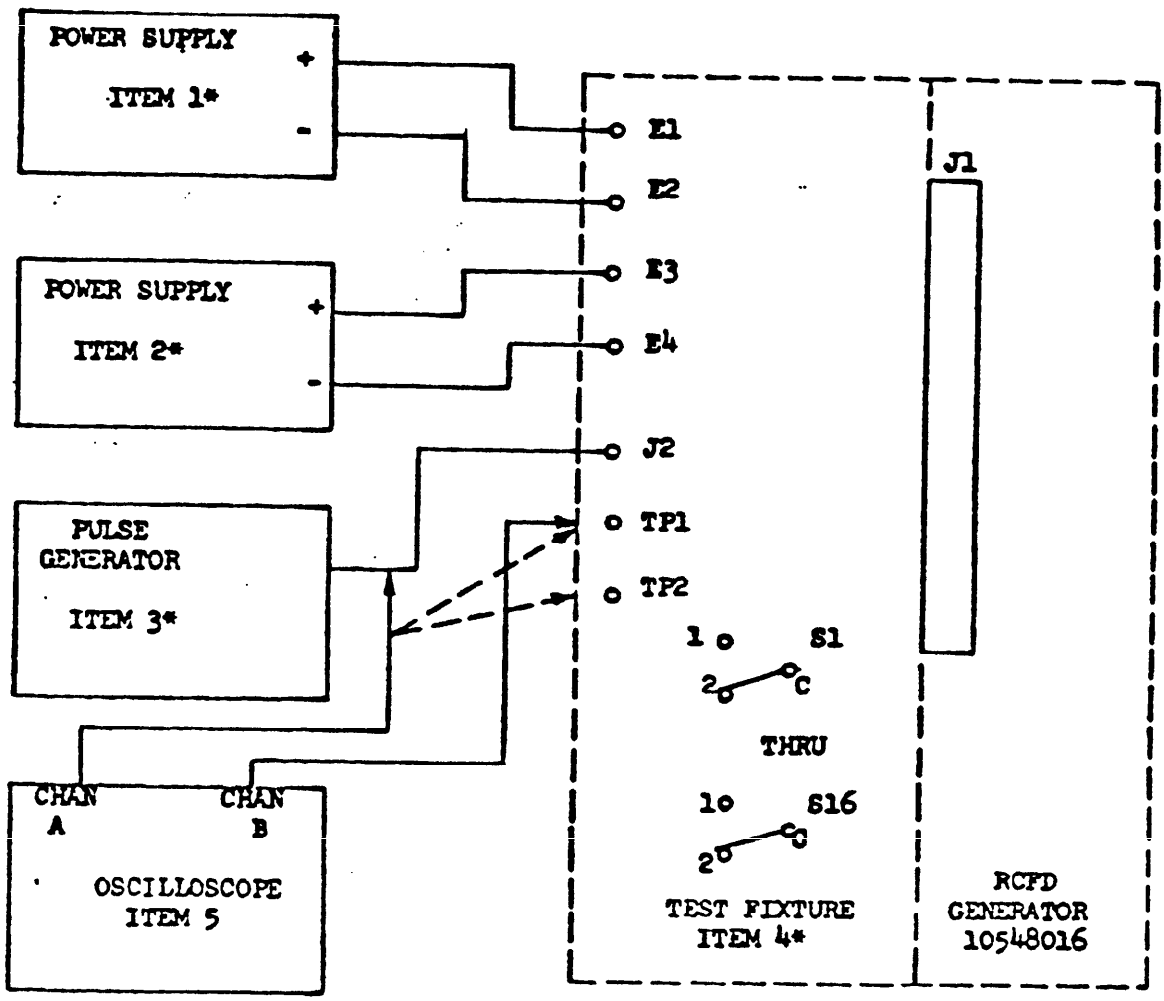
4.6.3.2 Initial delay. Using channels A and B of oscilloscope, item 5, to measure the input and output signal, the output pulse shall be as shown in Table 1 with a delay as specified in Step 1 of Table II.

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4.6.3.3 Gate decoder delays. Using charnel A of oscilloscope, the output pulse shall be as shown in Table I with the delays as specified in Table II for each Step 2 through 16.

4.7 Soldering. As specified in MIL-C-50652.

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*REFER TO 4.5.3

FIGURE 2. RCFD Generator Test Setup

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5. **PACKAGING**

5.1 **Packaging, packing and marking.** Packaging, packing and marking shall be in accordance with Packaging Data Sheet PS10548016. The level of protection shall be as specified in the procurement document.

6. **NOTES.** As specified in 6. Notes of MIL-C-50652.

**Custodian:**
Army-AR

**Preparing Activity:**
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(Project 1285-A775)

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3. DOCUMENT TITLE
CIRCUIT CARD ASSEMBLY; 10548016 (GENERATOR, RCFD)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

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